

Coil Design for Optimum Operating Range of Magnetically-Coupled RFID System

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Abstract — This paper describes a systematic design procedure of coils for optimal operating range of magnetically-coupled radio frequency identification (RFID) system. Given the sensitivities of both RFID reader and tag, the proposed design procedure, theoretically, calculates the inductance values and dimensions of both reader and tag coils, and predicts the corresponding optimal operating distance. Two design examples for both 125 kHz and 13.56 MHz RFID systems using typical sensitivity values are given to illustrate the proposed design procedure.

Keyword : RFID, Transponder, Inductive link, Coil design, Optimum inductance, Design procedure

I. INTRODUCTION

Radio-frequency identification system [1] has become the enabling technology in many applications, such as access control, transportation, traceability and wireless sensing. In many applications, it is desired to maximize the operating range of RFID. The operating range of RFID system depends on several factors including power transmitted by reader, energy coupling efficiency, power dissipation in the transponder, reader sensitivity, Q factor, data modulation. [2] General design guidelines for optimal read range of RFID system have been discussed in [1]-[4]. In practice, design and implementation of RFID system is built upon trial and error, cumulative experience of technicians and engineers. Therefore, one still lacks of a systematic design procedure for optimal operating range of RFID system.

This paper proposes a systematic design procedure for optimal operating range of RFID system. Given the sensitivities of both RFID reader and tag, the proposed design procedure, theoretically, calculates the inductance values and dimensions of both reader and tag coils, and predicts the corresponding optimal operating distance.

II. PROPOSED DESIGN PROCEDURE

Power and data transfers in a magnetically-coupled resonant RFID system can be equivalently represented as shown in Fig. 1. The reader front-end is represented by a current source driving a series resonant circuit (C_1 and L_1), which produces the maximum current through its coil at the resonant frequency. The transponder or tag front-end is represented by a parallel resonant circuit (C_2 and L_2), which produces a maximum voltage at its coil at the resonant frequency. Maximum coupling between both coils occurs when both resonant frequencies are identical, and this

resonant frequency is the carrier frequency of RFID system, which is typically 125 kHz or 13.56 MHz.

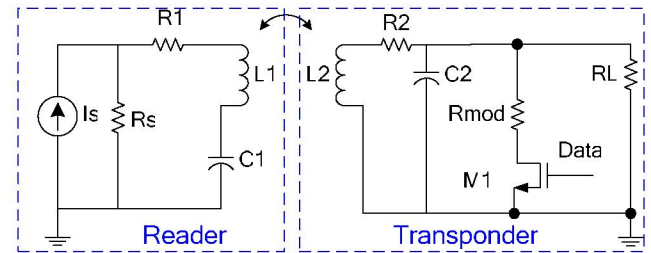


Fig. 1. Equivalent circuit of magnetically-coupled RFID system

The operating range (S_0) of a magnetically-coupled RFID system is determined by the lesser between (1) the maximum distance which the tag receives minimum required power for operation from the interrogating field of the reader (S_{R-T}), and (2) the maximum distance which the reader correctly detects the data sent from the tag (S_{T-R}).

S_{R-T} is determined by the RF power transmitted by reader, which is governed by regulations (e.g. FCC in the United States, ETSI in Europe, and MPT in Japan), and the minimum coil voltage (u_{2min}) required for transponder's operation. S_{T-R} is governed by the reader sensitivity and the modulation depth of the transponder [2], [3]. This paper describes a systematic design procedure of coils that optimizes the operating range of RFID system.

A. Optimum inductance of transponder's coil

In magnetically-coupled RFID system, load modulation is generally employed for data transmission from transponder to reader. Load modulation is performed by varying circuit parameters of the transponder resonant circuit in time with the transmitted data stream. There are generally two types of load modulation, namely resistive and capacitive. Load modulation changes the amplitude and phase of the reader resonant circuit through the transform transponder impedance (Z_T), allowing the transmitted data to be demodulated accordingly.

In the presence of load modulation, the reader's series resonant circuit can be represented as shown in Fig. 2. The voltage (u_0) developed at the reader's resonant circuit can be calculated as given in (1), where ω_0 is resonant frequency, M is the mutual inductance between transponder and reader coils, L_1 is the inductance of the reader's coil, L_2 is the

inductance of transponder's coil, i_1 is the current through in reader's coil, C_1 is the capacitance of the reader's resonant circuit, R_1 is the resistance of the reader's coil, R_2 is the resistance of the transponder's coil, and Z_L is the effective load impedance of transponder resonant circuit.

$$u_0 = j\omega_0 L_1 \cdot i_1 + \frac{1}{j\omega_0 C_1} + R_1 \cdot i_1 - j\omega_0 M \cdot \frac{j\omega_0 M i_1}{R_2 + j\omega_0 L_2 + Z_L} \quad (1)$$

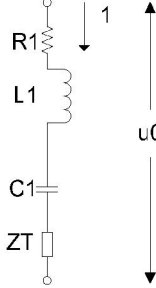


Fig. 2. Equivalent circuit of the reader series resonant circuit in the presence of load modulation

The impedance Z_L equals to R_L and $(R_L // R_{mod})$ when the transmitted data is '0' and '1', respectively. This changes the amplitude and phase of u_0 accordingly with the transmitted data. Changes in u_0 (Δu_0) due to resistive load modulation can be written as in (2). Maximum changes in u_0 allows data to be detected and demodulated more easily, and for a given reader sensitivity, it translates to longer RFID read range. This section analyzes the optimum inductance of transponder's coil to achieve the maximum change in u_0 .

$$\Delta u_0 = \left(\frac{\omega_0^2 k^2 L_1 L_2 i_1}{R_2 + j\omega_0 L_2 + R_L} - \frac{\omega_0^2 k^2 L_1 L_2 i_1}{R_2 + j\omega_0 L_2 + (R_L // R_{mod})} \right) \quad (2)$$

Given the values of R_L , R_{mod} , and R_2 , the optimum value of L_2 , which gives the maximum Δu_0 , can be calculated by differentiating (2) with respect to L_2 . The resulting optimum L_2 is given by (3). Fig. 3 plots Δu_0 against L_2 for different values of coupling coefficient k . It can be seen that the maximum Δu_0 can be obtained, for all values of k , by using the optimum L_2 . Fig. 4 also shows that, for a given reader sensitivity, the optimum L_2 produces the smallest k and thus the longest read range. It should also be noted that Δu_0 is linearly proportional with L_1 , as shown in (2).

$$L_2(opt) = \frac{\left[R_2 \cdot (R_L // R_{mod}) + R_L \cdot (R_L // R_{mod}) + R_2^2 + R_2 R_L \right]^{\frac{1}{2}}}{\omega_0} \quad (3)$$

The optimum L_2 can be used to calculate the dimension of the transponder's coil accordingly, for example in (4) for a circular coil [1], where N_2 is the number of turn, r_T is the radius, and d is the diameter.

$$L_2 = N_2^2 \cdot \mu_0 \cdot r_T \cdot \ln\left(\frac{2r_T}{d}\right) \quad (4)$$

B. Distance for energy transmission from reader to transponder (S_{R-T})

Using the optimum L_2 , the maximum distance for power transmission from reader to transponder can be calculated for a given minimum transponder's coil voltage (u_{2min}). According to [1], the distance for power transmission from reader to transponder (S_{R-T}) can be calculated by (5), where H_{min} , in (6), is the minimum interrogating magnetic field strength required for transponder's operation, and N_1 and r_R are the number of turns and radius of the reader's coil, respectively. Fig. 5 plots S_{R-T} against r_R , showing the existence of optimum r_R . For a given H_{min} , the optimum r_R can be calculated as given by (7). By substituting (7) into (5), the optimum S_{R-T} can be found. The optimum r_R is also used to design the reader's coil inductance according to the desired shape.

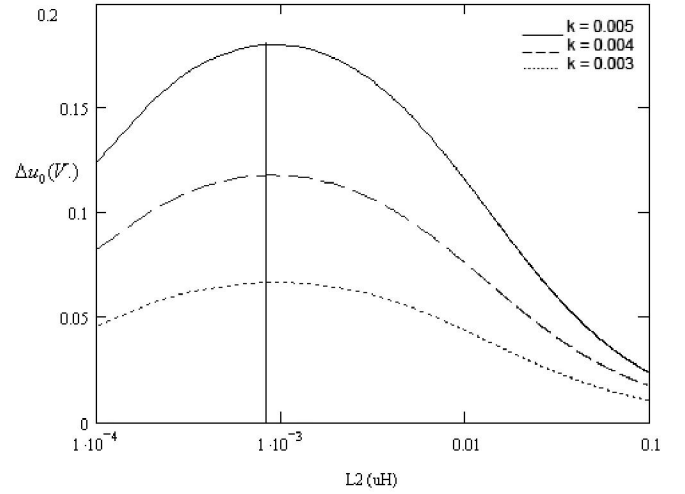


Fig. 3. Change in voltage of the reader's resonant circuit due to load modulation vs. inductance of the transponder's coil.

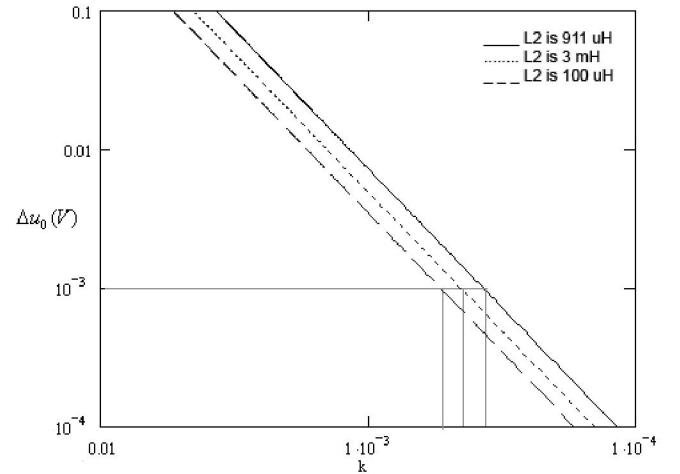


Fig. 4. Change in voltage of the reader's resonant circuit due to load modulation vs. coupling coefficient values.

$$S_{R-T} = \sqrt[3]{\left(\frac{i_1 \cdot N_1 \cdot r_R^2}{2 \cdot H_{\min}}\right)^2} - r_R^2 \quad (5)$$

$$H_{\min} = \frac{u_{2(\min)} \cdot \sqrt{\left(\frac{\omega_0 L_2}{R_L} + \omega_0 R_2 C_2\right)^2 + \left(1 - \omega_0^2 L_2 C_2 + \frac{R_2}{R_L}\right)^2}}{\omega_0 \cdot \mu_0 \cdot \pi \cdot r_T^2 \cdot N_2} \quad (6)$$

$$r_R(\text{opt}) = \frac{1}{9} \cdot \sqrt{6} \cdot N_1 \cdot \frac{i_1}{H_{\min}} \quad (7)$$

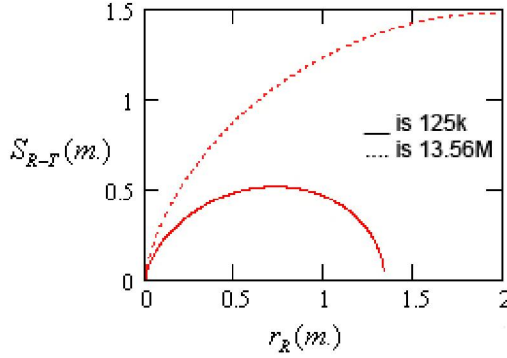


Fig. 5. Distance for energy transmission from reader to transponder

C. Distance for data transmission from transponder to reader (S_{T-R})

For a given reader's sensitivity ($\Delta u_{0\min}$), the minimum coupling coefficient ($k_{\min}$) that transponder can communicate with reader can be calculated by using (2), which is rearranged in (8) for convenience. Therefore the maximum distance for data transmission from transponder to reader (S_{T-R}) can be calculated by using (9).

$$k_{\min} = \sqrt{\Delta u_{0\min} \cdot \left(\frac{\omega_0^2 L_1 L_2 i_1}{R_2 + j\omega_0 L_2 + R_L} - \frac{\omega_0^2 L_1 L_2 i_1}{R_2 + j\omega_0 L_2 + (R_L // R_{\text{mod}})} \right)^{-1}} \quad (8)$$

$$k_{\min} = \frac{r_T^2 \cdot r_R^2}{\sqrt{r_T \cdot r_R} \cdot \left(\sqrt{S_{T-R}^2 + r_R^2} \right)^3} \quad (9)$$

Therefore the optimized operating range of RFID system is determined by the lesser between S_{R-T} and S_{T-R} given in (5) and (9), respectively. The aforementioned design procedure is summarized by the flow chart in Fig. 6.

III. DESIGN EXAMPLES

To illustrate the proposed design procedure, two design examples for 125 kHz and 13.56 MHz RFID systems are described in this section. Typical parameters in existing RFID system, summarized in Table I, are used for calculation. The results are summarized in Table II.

TABLE I
INPUT DESIGN PARAMETERS

Parameters	125 kHz	13.56 MHz
R_L (k Ω)	10	10
Z_L (Ω)	1	1
R_2 (Ω)	50	50
d (cm.)	0.1 (AWG 18) [4]	0.1 (AWG 18) [4]
N_1 (Turns)	40 [5]	4 [4]
N_2 (Turns)	100 [2]	3 [2]
$U_{2(\min)}$ (V.)	12	12
i_1 (A.)	1.11 [5]	1.11 [5]

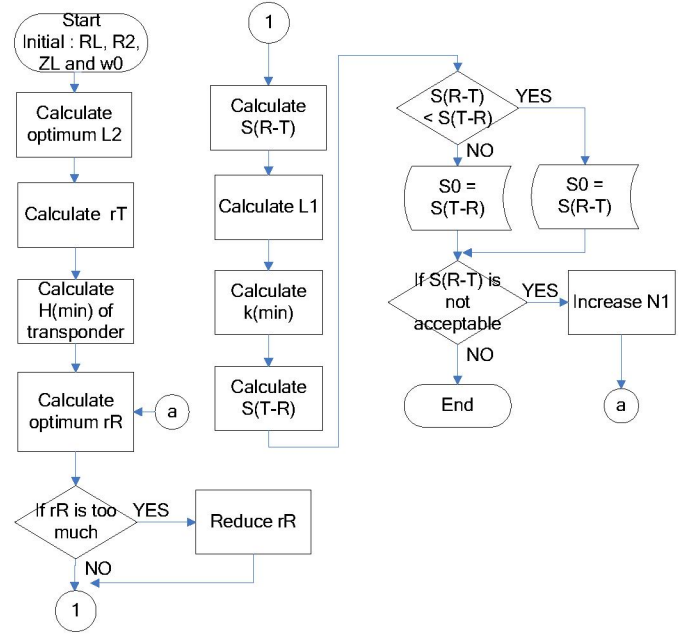


Fig. 6. Flowchart of the proposed design procedure.

TABLE II
RESULTS FROM THE DESIGN PROCEDURE.

Parameters	125 kHz	13.56 MHz
L_2 (uH)	911	8.4
L_1 (uH)	10,600	333
r_T (m)	0.02	0.018
r_R (m)	0.726	2
$H_{\min}$ (A/m)	15	0.55
S_{R-T} (m)	0.513	1.4
$k_{\min} (\Delta u_{0\min}=0.1V)$	0.0035	-
$k_{\min} (\Delta u_{0\min}=0.01V)$	0.0011	0.0006
$k_{\min} (\Delta u_{0\min}=1mV)$	-	0.0002
$S_{T-R} (\Delta u_{0\min}=0.1V)$ (m)	0.32	-
$S_{T-R} (\Delta u_{0\min}=0.01V)$ (m)	0.91	1
$S_{T-R} (\Delta u_{0\min}=1mV)$ (m)	-	2.55

Table II shows that S_{R-T} is equal to 0.513 m and 1.4 m for 125 kHz and 13.56 MHz systems, respectively. In the 125

kHz system, the maximum operating range (S_0) is equal to S_{T-R} of 0.32 m and S_{R-T} of 0.513 m when the reader's sensitivity (Δu_{0min}) is 0.1 V and 0.01V, respectively. In the 13.56 MHz system, S_0 is equal to S_{T-R} of 1 m and S_{R-T} of 1.4 m when the reader's sensitivity (Δu_{0min}) is 0.01 V and 0.001V, respectively. Therefore, it can be deduced that the maximum operating range of RFID system (S_0) is determined by S_{R-T} and S_{T-R} when the reader's sensitivity is high (i.e. Δu_{0min} is small) and low (i.e. Δu_{0min} is large), respectively.

Using the parameters in Tables I and II, Cadence Spectre simulations were performed to verify the results. Fig. 7 shows that the maximum change in u_0 is obtained at L_2 around 900 μH , as predicted in Table II. Fig. 9 plots Δu_0 against L_2 at different values of L_1 , which shows the optimum value of L_2 . It also shows that the optimum L_2 is independent of L_1 .

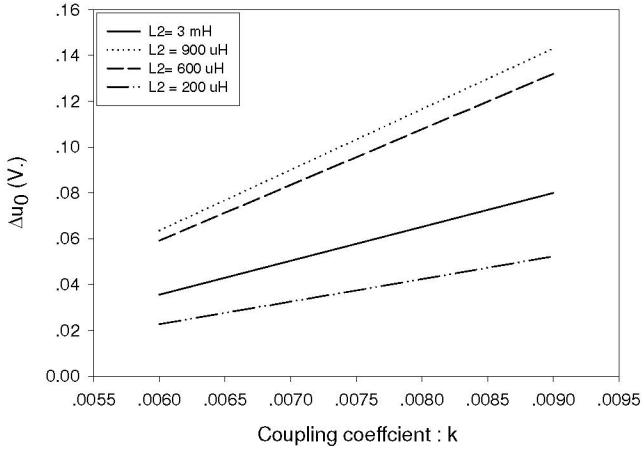


Fig. 7. Change in u_0 due to load modulation vs. k at different values of L_2 .

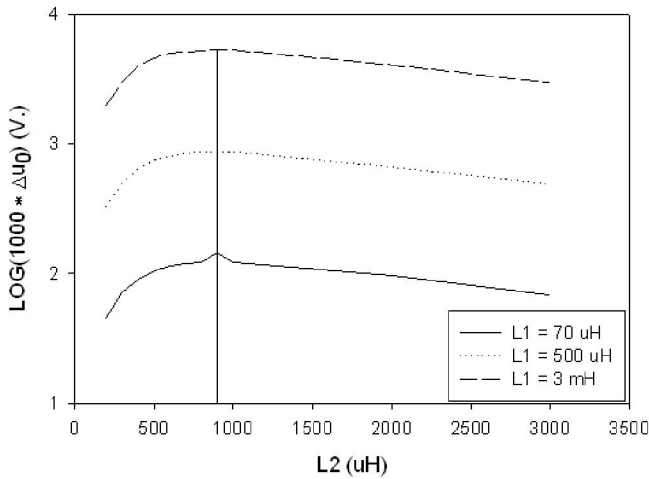


Fig. 8. Change in u_0 due to load modulation vs. L_2 at different values of L_1

IV. CONCLUSION

A systematic design procedure of coils for optimal operating range of magnetically-coupled RFID system has been described. Given the sensitivities of RFID reader and transponder, the proposed design procedure calculates the

optimum inductance values for both reader's coil and transponder's coil, and estimates the maximum operating distance of RFID system. The proposed procedure can be used to implement an interactive design tool for optimizing read range of RFID system.

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